□□□□□DeepUnfold□□□□□□□□□□□□

1. □□□□□□□□

- □□□□□□□□□□□□□□□□□□□□□□ LDL/Cholesky □□□□□□□□□□□□□□□□
- □□□□□M=256, K=32, batch=96□□□ ascend_910b_quiet □□□
- □□□□□ldl_256x32□deepunfold_256x32□deepunfold_opt_256x32□cholesky_256x32□
- trace □□□results/compare_aligned/*_trace.csv□

2. □□□□□□□□□□□□□□

□□□□□“□□□□”□□□□□□□□□□□□□□□

$$X_{k+1}=\mathcal{F}(X_k;A,\mathrm{Reg}),\quad k=0,1,\dots,L-1$$

- □□□□□□□□□□□□$A=H^H H + \mathrm{Reg}$□
- □□□□□□□□□□□□□□□□□□□□□□□□□□□□ $\hat{X}$□□
- □ NPU □□□□□MTE □□ + Cube(GEMM_PRELOAD) + Vector(ADD) □□□□□□

2.1 □□□□□□□□□□□□ NPU □□□□□

□□□□□□□□□□□□□□□□“□□□□□□□□□□□□□□□□□□□□□□”□NPU □□□□□□□□□□□□

1. □□□□□□□□□□ **MTE** □□□

- □□□□ X0 □□□□□□□□□□□□□□□□□□□□□□□□□□
- Z/□□□□□□□□□□□□□□□□□□□□ MOVIN/MOVOUT□
- □□□

2. □□□□□□□□□□□□□□□□□□

- □ DUO_VEC_CORR + DUO_VEC_MERGE □□□ DUO_VEC_UPDATE_Z□
- □□□□□□□□□□□□□□□□□□□□

□□□□□□□□□□□□□□

- `Gram□□□□□`□$A_g=H^H H$
- `□□□□`□$A=A_g+\lambda I$

2.3 □□□□□□□□□□□□□

□□□ A^{-1} □□□□□□□□□□□□□“□□”□□□□□□□□□□□□□□

2.6 □□□□□□□□□□□

□□□□□□□□□□□□□□□□□□□□ $\rho(I-AX_0)<1$□□□□□□□□□□□ A^{-1} □□□□□□□□□□□□□□□□□□□□□□□□□□□□ L□

$$ X_0 \rightarrow X_1 \rightarrow \cdots \rightarrow X_L $$

`DeepUnfold-Opt` □□□□□□□□□□□□□□□□□□□□

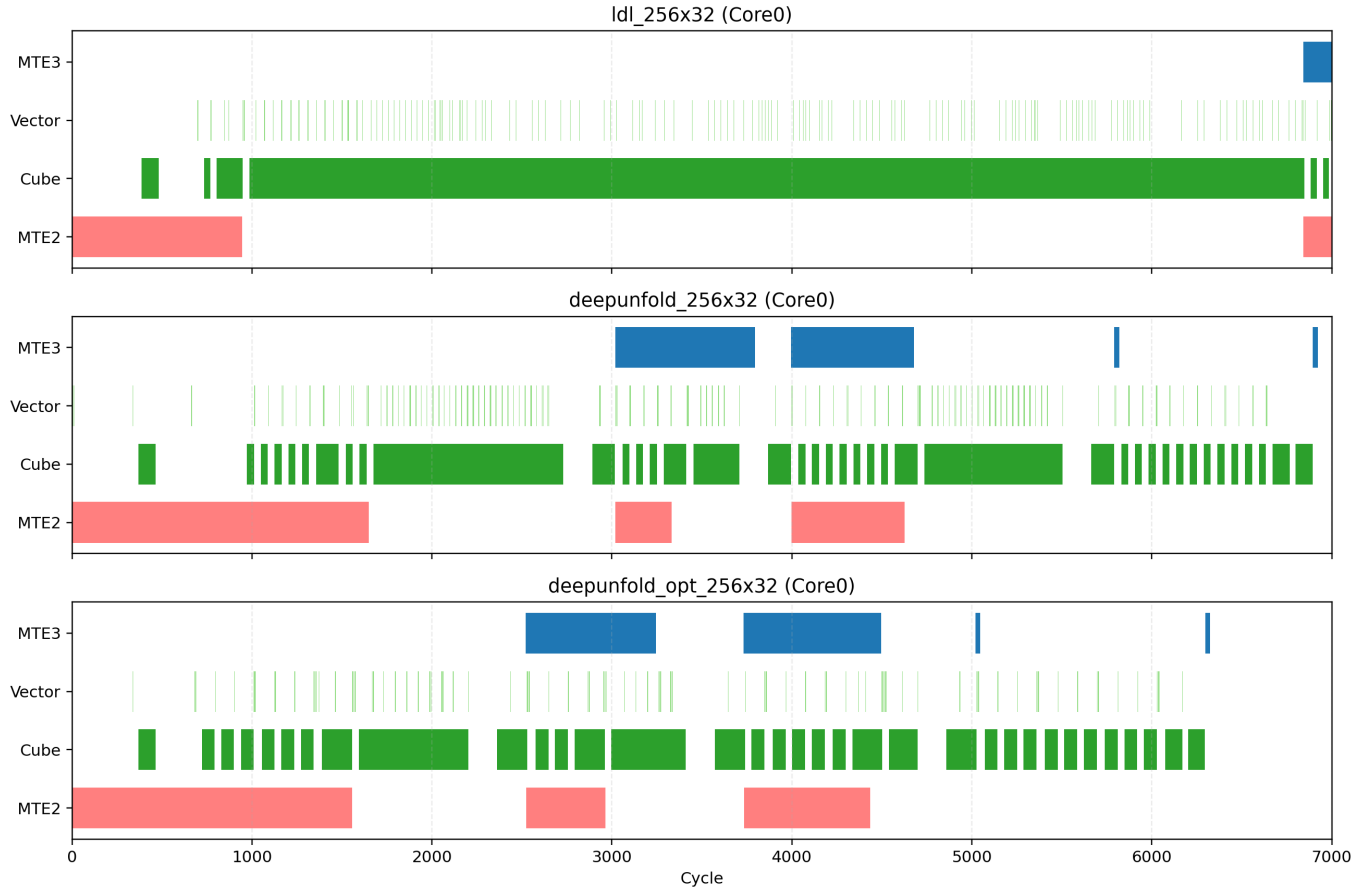
3. □□□□□□□□□

□□□□□□□□□□□□□□□□□□□□□□□□□□□□

- □□□□ `X0` □□□□□□□□□□□ `Xk`□
- □□ `Z`/□□□□□□□□□□□□□□□□□□□□□□□□□□
- □□□□□□□□□ `DUO_VEC_CORR + DUO_VEC_MERGE` □□□□□ `DUO_VEC_UPDATE_Z`□
- □□□□□□ `group_sync` barrier□□□□□□□□□□□□□□

> □□□□□□ `H/Y` □□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□□

4.1 □□□□□□□□□□□□□□□□□□□□□□□□



case	wall_cycles	work_cycles_sum	overlap_factor_sum_div_wall	wall_speed_vs_ldl
deepunfold_opt_256x32	6,370	29,321,742	4603.10	0.457
deepunfold_256x32	6,921	30,398,792	4392.25	0.496
ldl_256x32	13,953	12,487,766	894.99	1.000
cholesky_256x32	102,419	13,207,316	128.95	7.340

□4.1□□□□□□□□□□□

- □□□□□□□□□□□`deepunfold`/`deepunfold_opt` □□□□□□□□□ **LDL**□
- □□□□□□□“□□□□□”□□□ `deepunfold` □□□□□□□□□□□□□□`overlap_factor` □□□□
- □□□□□□□□□□□□□□□□□□□□`sum(dur)` □□□□□□`wall_cycles` □□□□□□

4.2 DeepUnfold □□□□□□□□□□ LDL □□□□□□□

□□□□□ LDL □□□□□

- □□□□`results/compare_aligned/deepunfold_256x32_trace.csv` □ `deepunfold_opt_256x32_trace.csv`

- □□□□□`dur = end_cycle - start_cycle`
- □□□□□`MTE2/MTE3 -> □□`□`Cube -> CUBE`□`Vector -> VECTOR`
- □□□□□□ `name` □□□□□□ `DU_AX_*`□`DU_XNEXT_*`□`DUO_VEC_UPDATE_Z_*`□

4.2.1 DeepUnfold□□□□□□□□□□□

Trace□`deepunfold_256x32_trace.csv`□`span=6921` cycles□

□□	□□	□□□□	**CUBE□□**	**VECTOR□□**	□□□	□□□
□□ (Load/Store)	$H/Y/X$ □□□□□	30,267,944	0	0	30,267,944	58,368
Gram□□□□	$A=H^H H$	0	9,120	192	9,312	288
□□□□	$A\leftarrow A+\lambda I$	0	0	384	384	96
□□□□□□ W	$W=f(A,H)$	0	9,120	0	9,120	96
AX□□□□	$r_k=A X_k$	0	44,928	0	44,928	1,152
□□/□□□□	$z_k=\phi(r_k,\cdots)$	0	0	4,608	4,608	1,152
X□□□□	$X_{k+1}=\psi(X_k,z_k)$	0	44,928	0	44,928	1,152
□□□□	$\hat X=\operatorname{head}(X_L)$	0	9,120	192	9,312	288
□□□□□□	X_k/Z_k □□□□□□	0	0	4,608	4,608	1,152
□□□□(Barrier)	□□/□□□□□□	0	0	3,648	3,648	3,648

4.2.2 DeepUnfold-Opt□□□□□□□□□□□

Trace□`deepunfold_opt_256x32_trace.csv`□`span=6370` cycles□

□□	□□	□□□□	**CUBE□□**	**VECTOR□□**	□□□	□□□
□□ (Load/Store)	$H/Y/X$ □□□□□	29,195,790	0	0	29,195,790	55,296
Gram□□□□	$A=H^H H$	0	9,120	192	9,312	288
□□□□	$A\leftarrow A+\lambda I$	0	0	384	384	96
□□□□□□ W	$W=f(A,H)$	0	9,120	0	9,120	96
□□□□□	$X_0\leftarrow \text{on-chip init}$	0	0	384	384	96
AX□□□□	$r_k=A X_k$	0	44,928	0	44,928	1,152
□□/□□□□	$z_k=\phi(r_k,\cdots)$	0	0	1,536	1,536	384
X□□□□	$X_{k+1}=\psi(X_k,z_k)$	0	44,928	0	44,928	1,152
□□□□	$\hat X=\operatorname{head}(X_L)$	0	9,120	192	9,312	288
□□□□□□	Z_k □□/□□□□	0	0	4,608	4,608	1,152
□□□□(Barrier)	□□/□□□□□□	0	0	1,440	1,440	1,440

4.2.3 □□□ DeepUnfold □□ Load/Store □□□□

- □□□□□□□□□□□□□□□□□□□□□□`MTE` □□□□ `99.57%`□□□4□□
- `DU/DUO` □□□□□□□□□□/□□□□□□□CUBE □□□□□□□□□□□□□□ `sum(dur)` □□□□□□□□□

- □□□□□□□□□ Cube □□“□”□□□□□□□□□□□□□□□□ Cube □□□□□□□□□ MTE □□□□□□□□□

4.3 LDL □□□□□□□□□□□□□□□□□

4.3.1 LDL vs DeepUnfold□aligned: `M=256,K=32,batch=96`□

□□□`results/compare_aligned/four_method_cycle_summary.csv` □ `four_method_wallclock_vs_work.csv`□

□□	LDL	DeepUnfold	DeepUnfold-Opt
Wall □□	13,953	6,921	**6,370**
Work □□□□`sum(dur)`□	12,487,766	30,398,792	29,321,742
MTE □□□	11,912,438	30,267,944	29,195,790
Cube □□	550,848	117,216	117,216
Vector □□	24,480	13,632	8,736
MTE □□	95.39%	99.57%	99.57%

□□□

- □□□□□□□□DeepUnfold □□□□ LDL□
- □□□□□□□□□DeepUnfold □□□□ LDL□□□□ MTE □□□
- □□□□□□□□□□□□□□□□□□□□□□□□

4.3.2 DeepUnfold □□□□□□□DeepUnfold vs DeepUnfold-Opt□

□□	□□□ DeepUnfold	□□□ DeepUnfold-Opt	□□
Wall □□	6,921	**6,370**	**-7.96%**
Work □□□□`sum(dur)`□	30,398,792	29,321,742	-3.54%
MTE □□□	30,267,944	29,195,790	-3.54%
Cube □□	117,216	117,216	0.00%
Vector □□	13,632	8,736	-35.92%

□□□□□□□□□□□□□□□□□□□□□□□□□□□Cube □□□□□□□□□□ Wall □□□□□□□□